

HW Week 10 Jones

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```
library(ggplot2)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v lubridate  1.9.4      v tibble     3.2.1
v purrr      1.0.4      v tidyr      1.3.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
ID <- c(1:6)
```

Create numeric vector 1 through 6

```
Race <- c("black", "white", "black", "hispanic", "white", "white")
```

Create character vector for race

```
class(Race) # Check object type of Race
```

```
[1] "character"
```

```
is.factor(Race) # Check if Race is a factor
```

```
[1] FALSE
```

```
is.character(Race) # Check if Race is a character vector
```

```
[1] TRUE
```

```
is.logical(Race) # Check if Race is logical
```

```
[1] FALSE
```

It is a character vector.

```
Voted_For_Obama <- c(TRUE, TRUE, TRUE, FALSE, FALSE, TRUE)
```

Create logical vector for Obama vote

```
class(Voted_For_Obama)
```

```
[1] "logical"
```

Check type of Voted_For_Obama

```
is.logical(Voted_For_Obama)
```

```
[1] TRUE
```

Confirm it is logical

It is a logical vector.

```
::: {.cell}

```{r .cell-code}
Party_ID <- c("Dem", "Dem", "Dem", "Rep", "Ind", "Rep")

:::
```

## Create party ID vector

```
Income_Level <- c("High", "Low", "Low", "High", "High", "Low")
```

## Create income level vector

```
Approval <- c(70, 80, 68, 20, 10, 60)
```

## Create numeric approval ratings

```
vote.data <- data.frame(ID, Race, Voted_For_Obama, Party_ID, Income_Level, Approval)
```

## Combine vectors into data frame

```
is.factor(vote.data$Race)
```

```
[1] FALSE
```

## Check if Race column became a factor in data frame

```
length(vote.data$Voted_For_Obama[vote.data$Voted_For_Obama == TRUE])
```

```
[1] 4
```

## Count TRUE (Obama voters)

```
vote.data$Voted_For_Obama[vote.data$Party_ID == "Ind"]
```

```
[1] FALSE
```

## Check if Independent voted for Obama

```
vote.white <- vote.data[vote.data$Race == "white",]
```

## Subset data to include only white respondents

```
vote.white[, 3]
```

```
[1] TRUE FALSE TRUE
```

## Show 3rd column (Voted\_For\_Obama) of subset

```
length(vote.white$Voted_For_Obama[vote.white$Voted_For_Obama == TRUE])
```

```
[1] 2
```

## Count TRUE in white subset

```
Country_ID <- seq(from = 1, to = 7)
Country_Name <- c("United States", "United Kingdom", "Japan", "China", "Brazil", "Germany",
Regime_Type <- c("Democracy", "Democracy", "Democracy", "Dictatorship",
 "Democracy", "Democracy", "Dictatorship")

GDP_PC <- c(51163, 39367, 46838, 6070, 11347, 41376, 3115)

EU_Member <- c(FALSE, TRUE, FALSE, FALSE, FALSE, TRUE, FALSE)

my.data <- data.frame(Country_ID, Country_Name, Regime_Type, GDP_PC, EU_Member)

my.data$Population <- c(318946000, 64105654, 127090000, 1367420000,
 203322000, 80781000, 87354300)

my.data$GDP <- my.data$GDP_PC * my.data$Population
Add population column

my.data$Is_Democracy <- my.data$Regime_Type == "Democracy"
Create logical column for democracies

my.data # Display full data frame
```

	Country_ID	Country_Name	Regime_Type	GDP_PC	EU_Member	Population
1	1	United States	Democracy	51163	FALSE	318946000
2	2	United Kingdom	Democracy	39367	TRUE	64105654
3	3	Japan	Democracy	46838	FALSE	127090000
4	4	China	Dictatorship	6070	FALSE	1367420000
5	5	Brazil	Democracy	11347	FALSE	203322000
6	6	Germany	Democracy	41376	TRUE	80781000
7	7	Egypt	Dictatorship	3115	FALSE	87354300
		GDP	Is_Democracy			
1	1.631823e+13		TRUE			
2	2.523647e+12		TRUE			
3	5.952641e+12		TRUE			
4	8.300239e+12		FALSE			
5	2.307095e+12		TRUE			
6	3.342395e+12		TRUE			
7	2.721086e+11		FALSE			